RECIPROCAL SINE PRODUCTS AND CENTERED EUCLIDEAN KERNELS

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ABSTRACT. For integers $n \geq 3$, consider

$$P(n) = \prod_{k=2}^{n-1} \sin\Big(\frac{\pi n}{k}\Big).$$

We record exact residue, gamma, Fourier, divisor-indexed, and Euclidean forms of this varying-denominator product. The elementary block argument for the fractional parts $\{n/k\}$ gives the limiting density $\rho(t) = \psi_1(1+t)$ and the scale $\log|P(n)| = -Cn + O(\sqrt{n}\log n)$ on the nonzero locus. Specializing a theorem of Saffari–Vaughan to the diagonal range improves the uniform counting discrepancy from $O(\sqrt{n})$ to $O(n^{1/3}\log n)$; after the logarithmic endpoint singularities are retained explicitly this gives, for primes,

$$\log|P(p)| = -Cp + O\big(p^{1/3}(\log p)^2\big).$$

The same input sharpens the zero-omitted product and its primitive divisor component, and moves the half-plane of absolute convergence of the associated Dirichlet series from $\Re s > 3/2$ to $\Re s > 4/3$.

For prime arguments the reciprocal Fourier modes admit an exact centering, and their partial sums are a single Dirichlet-kernel quadrature defect. The exact Parseval identity for the rotated logarithmic product now yields $\|\mathscr{R}_p\|_2 \ll p^{1/3}\log p$. More specifically, the unrotated remainder sees only the folded discrepancy $H_p(t) = D_p(t) - D_p(1-t)$. We prove the exact formula

$$\log|P(p)| + Cp - 2\log 2 = -\pi \int_0^{1/2} H_p(t)\cot(\pi t)\,dt,$$

show that the extreme singular range $t \leq p^{-2/3}$ contributes only $O(p^{1/3}\log p)$, using the current Li–Yang divisor bound, and isolate the remaining logarithmic loss to the mesoscopic interval $p^{-2/3} \leq t \leq 1/2$. An exact regular-grid representation of H_p as a symmetric difference of reciprocal sawtooth sums is also proved, with all endpoint corrections displayed.

Finally, writing $s_k = k\|p/k\|$ separates the singular logarithm into a smooth bounded-variation term and an integer core $S_p = \prod s_k$. The core has exact shifted-divisor and gamma factorizations; the corrected gamma formula is verified numerically and a uniform Stirling expansion shows that the complementary-divisor sector $q \leq p^{1/3}$ can be smoothed at total cost $O(p^{1/3}\log p)$. Extensive numerical audits are included, in particular exhaustive one-sided jump maxima for the folded discrepancy. The observed bound $\max|H_p| \asymp p^{1/3}$ is stated only as a numerical conjecture. The finite encodings of π, μ, Λ, and Ψ remain exact reorganizations of classical arithmetic data and do not sharpen the von Koch criterion for the Riemann hypothesis.

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1. Introduction

This paper studies the diagonal, varying-denominator product

$$P(n)=\prod_{k=2}^{n-1}\sin\left(\frac{\pi n}{k}\right). \tag{1.1}$$

It differs from the classical complete product with a fixed denominator and from Sudler products, in which the frequency is fixed while the number of factors grows. Fixed-denominator products, partial rational sine products, and Sudler products have extensive theories; see [15, 10, 3]. The distribution of fractional parts of x/n is likewise classical [12]. For the specific diagonal family $\{n/k\}$, the same explicit pushforward density used below was also written out in a public 2023 derivation [11].

The aim here is structural and quantitative. We derive exact identities for (1.1), first by an elementary block argument and then by importing the sharper diagonal discrepancy estimate of Saffari and Vaughan. The latter improves the proved logarithmic remainder to $O(p^{1/3}(\log p)^2)$ on prime arguments. The product-specific step is then to exploit the reflection symmetry of the logarithmic sine kernel: the unrotated remainder depends only on a folded discrepancy, and its extreme singular part can be treated through symmetric shifted-divisor counts. We also isolate an integer core and a complementary-divisor gamma factorization, retaining all floor and residue terms until their total cost is proved. Separately, the Euclidean Lambert kernel, the floor-sensitive Clausen remainder, and the exact finite encodings of classical arithmetic functions are retained from the earlier development. Whenever a calculation reaches an established theorem or identity, that

input is stated with a reference. In particular, the appearances of μ, Λ, ζ, and Ψ below do not furnish new information about their size or analytic continuation.

2. Exact forms

For $n \geq 3$, write

$$m_k = \left\lfloor \frac{n}{k} \right\rfloor, \qquad r_k = n \bmod k, \qquad 0 \leq r_k < k. \tag{2.1}$$

2.1. **Residues and sign.** The identity

$$\sin\left(\frac{\pi n}{k}\right) = (-1)^{m_k} \sin\left(\frac{\pi r_k}{k}\right) \tag{2.2}$$

separates the sign from the nonnegative residue factor.

Proposition 2.1. *For every integer $n \geq 3$,*

$$P(n) = (-1)^{n+1+\lfloor\sqrt{n}\rfloor} \prod_{k=2}^{n-1} \sin\left(\frac{\pi(n \bmod k)}{k}\right). \tag{2.3}$$

For odd n, the sign exponent reduces modulo two to $\lfloor\sqrt{n}\rfloor$.

Proof. The sign exponent before reduction is

$$\sum_{k=2}^{n-1} \left\lfloor \frac{n}{k} \right\rfloor = D(n) - n - 1, \qquad D(n) = \sum_{j \leq n} d(j).$$

Since $d(j)$ is odd exactly when j is a square, $D(n) \equiv \lfloor\sqrt{n}\rfloor \pmod 2$. Formula (2.3) follows. □

Taking absolute values of roots of unity gives

$$P(n) = (-1)^{n+1+\lfloor\sqrt{n}\rfloor} 2^{-(n-2)} \prod_{k=2}^{n-1} \left|1 - e^{2\pi i n/k}\right|. \tag{2.4}$$

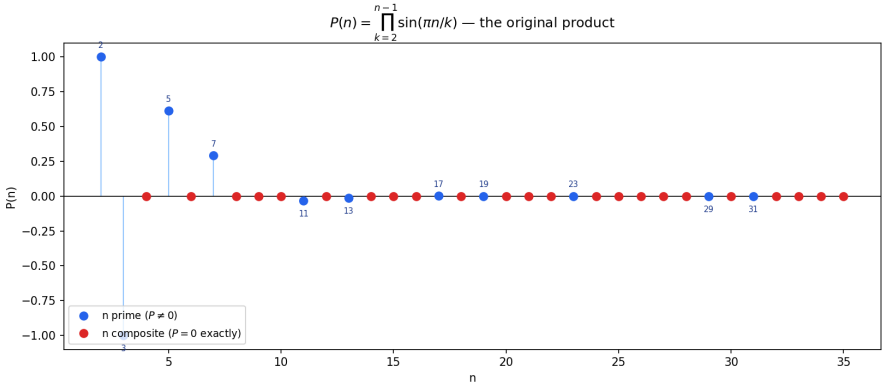


Figure 1. The product $P(n)$ itself, with sign, for $3 \leq n \leq 35$. Composite n give the exact value 0; prime n give a nonzero value of rapidly decreasing magnitude and no fixed sign pattern. The exponential decay rate is quantified in Theorem 3.5.

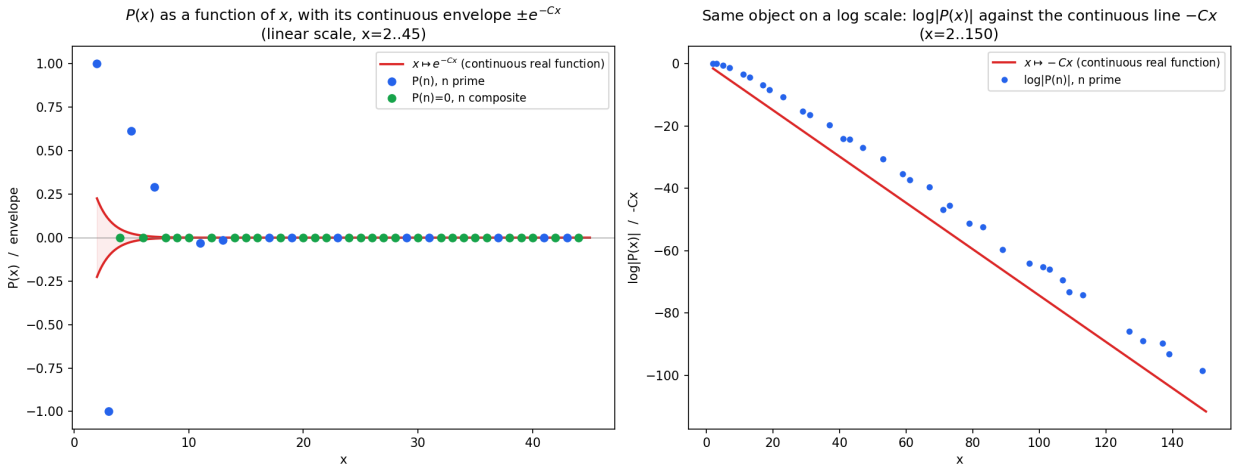


FIGURE 2. $P(n)$ treated explicitly as a function of a real variable x, by comparison against the genuine continuous function $x \mapsto e^{-Cx}$ (left, linear scale, $x \le 45$) and $x \mapsto -Cx$ (right, logarithmic scale, $x \le 150$). The discrete values $P(n)$ for prime n lie inside the continuous envelope $\pm e^{-Cx}$; composite n sit on the axis. Both panels display the same first-order law of Theorem 3.2 as an explicit function of x, not merely as a discrete sequence.

2.2. **Gamma, Fourier, and shifted-divisor forms.** Euler's reflection formula gives, whenever every factor is nonzero,

$$P(n) = \frac{\pi^{n-2}}{\prod_{k=2}^{n-1} \Gamma(n/k)\Gamma(1-n/k)}. \tag{2.5}$$

The Abel-regularized Fourier series of $\log|2\sin \pi t|$ gives

$$\log|P(n)| = -(n-2)\log 2 - \lim_{u\uparrow 1} \sum_{j\ge 1} \frac{u^j}{j} \sum_{k=2}^{n-1} \cos\left(\frac{2\pi j n}{k}\right). \tag{2.6}$$

The inner sums are reciprocal-phase exponential sums of the type appearing in the Dirichlet divisor problem; see [2].

For a nonzero residue $r = n \bmod k$, one has $k \mid n-r$ and $k > r$, and the converse is immediate. Hence

$$|P(n)| = \prod_{r=1}^{n-2} \prod_{\substack{k \mid n-r \\ k>r}} \sin\left(\frac{\pi r}{k}\right) \tag{2.7}$$

whenever $P(n) \neq 0$. More locally,

$$r_k = s \iff k \mid n-s, \qquad k - r_k = s \iff k \mid n+s. \tag{2.8}$$

3. Block distribution and exponential scale

Partition the indices by $m = \lfloor n/k \rfloor$ and $t = \{n/k\}$. The block m corresponds to $n/(m+1) < k \le n/m$. Here ψ_0 and ψ_1 denote the di- and trigamma functions. The distribution of fractional parts of x/n belongs to the classical framework of Saffari and Vaughan [12]. For the exact family $\{n/k\}$, the interval decomposition and the resulting density $\sum_{m\ge 1}(m+t)^{-2}$ were also displayed explicitly in the 2023 public derivation [11]. The following short proof records the uniform estimate needed here.

Theorem 3.1. *Uniformly for* $0 \le t \le 1$,

$$\#\left\{2 \le k \le n-1 : \left\{\frac{n}{k}\right\} \le t\right\} = n(\psi_0(1+t) - \psi_0(1)) + O(\sqrt{n}). \tag{3.1}$$

Consequently, for every continuous $f : [0,1] \to \mathbb{C}$,

$$\frac{1}{n}\sum_{k=2}^{n-1} f\left(\left\{\frac{n}{k}\right\}\right) \longrightarrow \int_0^1 f(t)\rho(t)\,\mathrm{d}t, \qquad \rho(t) = \psi_1(1+t) = \sum_{m\ge 1}\frac{1}{(m+t)^2}. \tag{3.2}$$

The density has total mass one.

Proof. Within block m, the additional condition $\{n/k\} \le t$ selects an interval containing

$$\frac{n}{m} - \frac{n}{m+t} + O(1) = \frac{nt}{m(m+t)} + O(1)$$

integers. Summing for $m \le M$, and bounding both the omitted indices and the main-term tail by $O(n/M)$, gives total error $O(M + n/M)$. Choose $M = \lfloor\sqrt{n}\rfloor$ and use

$$\sum_{m\ge 1}\frac{t}{m(m+t)} = \sum_{m\ge 1}\left(\frac{1}{m} - \frac{1}{m+t}\right) = \psi_0(1+t) - \psi_0(1).$$

Weak convergence gives (3.2), and differentiation gives the density. Its mass is $\sum_{m\ge 1}(1/m - 1/(m+1)) = 1$. □

Collapse principle. For every continuous $F : [0,1] \to \mathbb{C}$,

$$\frac{1}{n}\sum_{k=2}^{n-1} F\left(\left\{\frac{n}{k}\right\}\right) \longrightarrow \int_0^1 F(t)\,\psi_1(1+t)\,\mathrm{d}t. \tag{3.3}$$

This is exactly the weak-convergence statement (3.2): writing $u = k/n$, the grid points become Lebesgue distributed on $(0,1]$ and $\{n/k\} = \{1/u\}$. Hence the diagonal geometry fixes a single limiting pushforward class. The observable F changes the tested integral, but not the underlying measure.

The principle explains why several apparently different constructions in this paper return to the same object. The residue, gamma, Fourier, and shifted-divisor formulas of Section 2 are exact representations of the same diagonal observable; for $F(t) = \log|\sin \pi t|$, the endpoint singularities require the separate truncation estimate in Theorem 3.2, but the limiting measure is unchanged. At the other end of the paper, the proof of Proposition 9.4 again uses $g(u) = \mathrm{Cl}_2(2\pi\{1/u\})$, now with totient weights. Accordingly, changing the kernel changes the tested observable and its arithmetic weights, not the underlying fractional-part pushforward class.

The measure in (3.2) is the one-step pushforward of Lebesgue measure by the Gauss map $u \mapsto \{1/u\}$. Its block masses are $1/(m(m+1))$, not the invariant Gauss–Kuzmin digit law; compare [9]. For example, the limiting proportion of negative sine factors is

$$\sum_{\substack{m\ge 1\\ m \text{ odd}}}\frac{1}{m(m+1)} = \log 2. \tag{3.4}$$

The logarithmic singularities at the endpoints require a separate estimate.

Theorem 3.2. *Along integers for which* $P(n) \ne 0$,

$$\log|P(n)| = -Cn + O(\sqrt{n}\log n), \tag{3.5}$$

where

$$C = -\int_0^1 \log(\sin \pi t)\, \psi_1(1+t)\, \mathrm{d}t \tag{3.6}$$

$$= -\int_1^\infty \frac{\log|\sin \pi u|}{u^2}\, \mathrm{d}u = -\int_0^1 \log\left|\sin \frac{\pi}{u}\right|\, \mathrm{d}u = 0.7437948618\ldots. \tag{3.7}$$

Proof. For a nonzero factor, $\sin(\pi r_k/k) \geq \sin(\pi/k) \geq 2/k$. Thus the indices $k \leq \sqrt{n}$ contribute $O(\sqrt{n}\log n)$.

For $k > \sqrt{n}$, there are fewer than $\sqrt{n}$ blocks. Within each block, $k \mapsto \{n/k\}$ is monotone and $t \mapsto \log\sin(\pi t)$ has two monotone pieces. Comparing the sum with its integral therefore costs $O(\log n)$ per block after the singular endpoints are trimmed at the nearest admissible sample. The aggregate error is $O(\sqrt{n}\log n)$.

With $u = n/x$, the main integral is

$$n\int_{1+1/(n-1)}^{\sqrt{n}} \frac{\log|\sin \pi u|}{u^2}\, \mathrm{d}u.$$

Extending the upper limit to infinity costs $O(\sqrt{n})$. At the lower endpoint the omitted integral is

$$n\int_1^{1+1/(n-1)} \frac{\log|\sin \pi u|}{u^2}\, \mathrm{d}u = -\log n + O(1),$$

which is absorbed by the stated $O(\sqrt{n}\log n)$ remainder. This proves (3.5). The substitutions $u = m + t$ and $u \mapsto 1/u$ give (3.6)–(3.7). □

Remark 3.3 (Endpoint contribution). The lower-end calculation in the proof isolates a contribution $+\log n + O(1)$ when the main integral is extended down to $u = 1$. Since the proven global error is $O(\sqrt{n}\log n)$, this observation is not asserted as a second-order asymptotic term. In particular, Theorem 3.2 claims only the first-order law $-Cn$ with the stated remainder.

3.1. **Saffari–Vaughan refinement of the diagonal discrepancy.** The preceding proof is elementary but not quantitatively optimal. The standard case of Saffari–Vaughan [12, Theorem 1 and Corollary 1.1] contains the sharper diagonal estimate needed here.

Theorem 3.4 (Saffari–Vaughan specialization)**.** *Uniformly for* $0 \leq t \leq 1$,

$$\#\left\{2 \leq k \leq n-1 : \left\{\frac{n}{k}\right\} \leq t\right\} = nF(t) + O\left(n^{1/3}\log n\right), \qquad F(t) = \psi_0(1+t) - \psi_0(1). \tag{3.8}$$

The choice of strict or non-strict endpoint convention changes the count by at most the multiplicity of the corresponding jump and does not alter the stated estimate.

Proof. Saffari and Vaughan normalize the count by the length y. In their Theorem 1 the error is $O(x^{1/3}y^{-1}\log x)$. Set $x = y = n$ and use their Corollary 1.1, whose limiting function is

$$\sum_{m\geq 1} \frac{t}{m(m+t)} = \psi_0(1+t) - \psi_0(1) = F(t).$$

Multiplying the normalized estimate by n gives (3.8); removing the two endpoints $k = 1, n$ costs only $O(1)$. □

Theorem 3.5 (Improved prime logarithmic scale)**.** *For every prime* p,

$$\log|P(p)| = -Cp + O\left(p^{1/3}(\log p)^2\right). \tag{3.9}$$

Proof. Put $g(t) = \log|2\sin \pi t|$, and let $E_p \ll p^{1/3}\log p$ denote the uniform discrepancy supplied by Theorem 3.4. Choose $\delta = p^{-2/3}$. On $[\delta, 1-\delta]$, Stieltjes integration by parts gives an error

$$O\left(E_p \operatorname{Var}_{[\delta,1-\delta]} g\right) = O\left(p^{1/3}(\log p)^2\right),$$

because $g'(t) = \pi \cot(\pi t)$ and the truncated variation is $O(\log(1/\delta))$.

At either endpoint, Theorem 3.4 gives $O(p\delta + E_p)$ sample points. Primality implies $\|p/k\| \geq 1/k \geq 1/p$, so every sampled logarithm is $O(\log p)$ in absolute value. The corresponding endpoint integrals are $O(p\delta \log p)$. These contributions are again $O(p^{1/3}(\log p)^2)$. Finally,

$$\sum_{k=2}^{p-1} g(p/k) - p \int_0^1 g(t)\rho(t)\, dt = \log|P(p)| + Cp - 2\log 2,$$

which proves the result. □

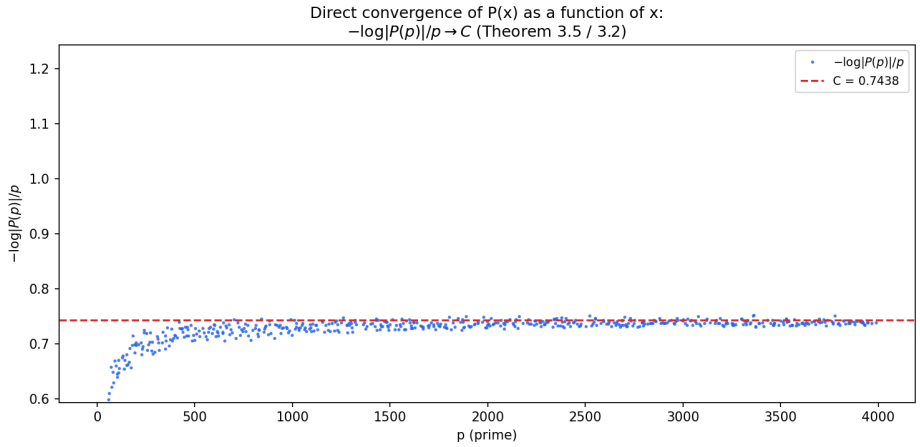


FIGURE 3. Direct numerical illustration of Theorem 3.5: $-\log|P(p)|/p$ plotted against p for every prime $p < 4000$, converging to the constant $C = 0.7437948618\ldots$ of Theorem 3.2. This is the same convergence stated in (3.9), displayed without passing through a logarithmic axis.

The same constant has the convergent representation

$$C = \log 2 + \sum_{j\geq 1} \frac{1}{j}\left[1 - 2\pi j\left(\frac{\pi}{2} - \mathrm{Si}(2\pi j)\right)\right]. \tag{3.10}$$

Proposition 3.6. *The constant C satisfies $C > \log 2$.*

Proof. Put $g(t) = \log(2\sin \pi t)$. Then $C - \log 2 = -\sum_{m\geq 1} I_m$, where

$$I_m = \int_0^{1/2} g(t)\big((m+t)^{-2} + (m+1-t)^{-2}\big)\, \mathrm{d}t.$$

On $(0, 1/2)$, g is strictly increasing, the factor multiplying it is strictly decreasing, and $\int_0^{1/2} g(t)\, \mathrm{d}t = 0$. The reversed Chebyshev integral inequality gives $I_m < 0$; see [4]. Summing proves the assertion. □

4. Centered reciprocal frequencies and quantitative rotated bounds

For prime arguments, the reciprocal Fourier modes can be centered against the same density before the singular Fourier series is summed.

4.1. **Centered modes and the exact prime remainder.** For $j \geq 1$, define

$$c_j = \int_0^1 \cos(2\pi jt)\rho(t)\,dt, \qquad \rho(t) = \psi_1(1+t). \tag{4.1}$$

Periodicity and the block expansion of ρ give

$$c_j = \int_1^\infty \frac{\cos(2\pi ju)}{u^2}\,du = 1 - 2\pi j\left(\frac{\pi}{2} - \mathrm{Si}(2\pi j)\right), \tag{4.2}$$

and repeated integration by parts gives

$$c_j = \frac{1}{2\pi^2 j^2} - \frac{3}{2\pi^4 j^4} + O(j^{-6}). \tag{4.3}$$

Thus

$$C = \log 2 + \sum_{j\geq 1} \frac{c_j}{j}. \tag{4.4}$$

For a prime p, put

$$S_j(p) = \sum_{k=2}^{p-1} \cos\left(\frac{2\pi jp}{k}\right), \qquad \Delta_j(p) = S_j(p) - pc_j. \tag{4.5}$$

Theorem 4.1 (Exact centered Fourier remainder)**.** *For every prime p,*

$$\log|P(p)| + Cp = 2\log 2 - \lim_{u\uparrow 1} \sum_{j\geq 1} \frac{u^j}{j}\Delta_j(p). \tag{4.6}$$

Proof. Insert $S_j(p) = pc_j + \Delta_j(p)$ into (2.6) and use (4.4). The Abel regularization is unchanged from (2.6). □

4.2. **A single quadrature defect for the cumulative modes.** Set

$$B_J(p) = \sum_{j=1}^{J} \Delta_j(p), \qquad K_J(t) = \sum_{j=1}^{J} \cos(2\pi jt). \tag{4.7}$$

Proposition 4.2 (Exact reciprocal quadrature identity)**.** *For every prime p and every integer $J \geq 1$,*

$$B_J(p) = \sum_{k=2}^{p-1} K_J\left(\frac{p}{k}\right) - \int_0^p K_J\left(\frac{p}{x}\right)\,dx, \tag{4.8}$$

where the integral at $x = 0$ is improper.

Proof. By definition,

$$B_J(p) = \sum_{k=2}^{p-1} K_J(p/k) - p\int_0^1 K_J(t)\rho(t)\,dt.$$

Since $\rho(t) = \sum_{m\geq 1}(m+t)^{-2}$ and K_J is one-periodic,

$$\int_0^1 K_J(t)\rho(t)\,dt = \int_1^\infty \frac{K_J(u)}{u^2}\,du.$$

The substitution $x = p/u$ gives (4.8). □

Remark 4.3. The quadrature identity is exact, but a formerly considered bound $|B_J(p)| \ll \sqrt{pJ}$ would only imply an $O(\sqrt{p})$ pointwise remainder. Theorem 3.5 is asymptotically stronger, so such a square-root criterion is no longer the quantitative target of this paper. In particular, no complete-period average in the Fourier index is used to infer a short-frequency estimate.

4.3. **Exact energy identity and the strengthened rotated theorem.** For a prime p, define

$$N_p(t) = \#\left\{2 \le k \le p-1 : \left\{\frac{p}{k}\right\} \le t\right\}, \qquad D_p(t) = N_p(t) - pF(t) + 2t. \tag{4.9}$$

Then $D_p(0) = D_p(1) = 0$, and Theorem 3.4 gives

$$\|D_p\|_\infty \ll p^{1/3}\log p. \tag{4.10}$$

For $j \ge 1$, put

$$A_j(p) = \sum_{k=2}^{p-1} e^{2\pi i j p/k} - p\int_0^1 e^{2\pi i j t}\rho(t)\,dt. \tag{4.11}$$

Lemma 4.4 (Exact energy identity)**.** *With $\overline{D}_p = \int_0^1 D_p(t)\,dt$,*

$$\sum_{j\ge1}\frac{|A_j(p)|^2}{j^2} = 2\pi^2\int_0^1 |D_p(t) - \overline{D}_p|^2\,dt \ll p^{2/3}(\log p)^2. \tag{4.12}$$

Proof. The signed measure

$$d\mu_p(t) = \sum_{k=2}^{p-1}\delta_{\{p/k\}}(dt) - p\rho(t)\,dt + 2\,dt$$

has cumulative function D_p and total mass zero. Thus

$$A_j(p) = -2\pi i j\int_0^1 D_p(t)e^{2\pi i j t}\,dt.$$

Parseval applied to $D_p - \overline{D}_p$ proves the identity; the final bound follows from (4.10). □

For $\theta \in \mathbb{R}/\mathbb{Z}$, define

$$\mathscr{R}_p(\theta) = \sum_{k=2}^{p-1}\log\left|2\sin\pi\left(\frac{p}{k}+\theta\right)\right| - p\int_0^1 \log|2\sin\pi(t+\theta)|\rho(t)\,dt. \tag{4.13}$$

Theorem 4.5 (Quantitative rotated L^2 bound)**.** *For every prime p,*

$$\int_0^1 |\mathscr{R}_p(\theta)|^2\,d\theta = \frac{1}{2}\sum_{j\ge1}\frac{|A_j(p)|^2}{j^2} = \pi^2\int_0^1 |D_p(t)-\overline{D}_p|^2\,dt \ll p^{2/3}(\log p)^2. \tag{4.14}$$

Moreover,

$$\mathscr{R}_p(0) = \log|P(p)| + Cp - 2\log 2. \tag{4.15}$$

Proof. The Fourier coefficients of $g(t) = \log|2\sin\pi t|$ are $\widehat{g}(j) = -1/(2|j|)$ for $j \ne 0$. Hence $\widehat{\mathscr{R}_p}(j) = -A_j(p)/(2j)$ for $j \ge 1$, and Parseval with Lemma 4.4 proves (4.14). The value at $\theta = 0$ follows from $\int_0^1 g(t)\rho(t)\,dt = -C + \log 2$. □

Corollary 4.6. *For every $\lambda > 0$,*

$$\operatorname{meas}\left\{\theta : \ |\mathscr{R}_p(\theta)| > \lambda p^{1/3}\log p\right\} \ll \lambda^{-2}. \tag{4.16}$$

Proof. Apply Chebyshev's inequality to (4.14). □

5. Folded discrepancy and localization of the logarithmic loss

The logarithmic sine kernel is reflection invariant, so the unrotated remainder does not see the full discrepancy D_p.

Definition 5.1. For $0 \le t \le 1/2$, put

$$H_p(t) = D_p(t) - D_p(1-t). \tag{5.1}$$

Also define

$$M_p(t) = \#\left\{2 \le k \le p-1 : \left\|\frac{p}{k}\right\| \le t\right\}, \qquad F_{\mathrm{fold}}(t) = 1 + F(t) - F(1-t). \tag{5.2}$$

Away from the finitely many boundary points, and with the corresponding one-sided convention at a boundary,

$$H_p(t) = M_p(t) - pF_{\mathrm{fold}}(t) + 4t. \tag{5.3}$$

In particular $H_p(0) = H_p(1/2) = 0$.

Theorem 5.2 (Exact folded Stieltjes formula)**.** *For every prime p,*

$$\log|P(p)| + Cp - 2\log 2 = -\pi \int_0^{1/2} H_p(t)\cot(\pi t)\,dt, \tag{5.4}$$

where the integral is improper at zero.

Proof. With $g(t) = \log|2\sin \pi t|$, the left side is $\int_0^1 g\,dD_p$. Since $D_p(0) = D_p(1) = 0$, Stieltjes integration by parts gives

$$-\pi \int_0^1 D_p(t)\cot(\pi t)\,dt.$$

On the second half substitute $t \mapsto 1-t$ and use $\cot(\pi(1-t)) = -\cot(\pi t)$. The two halves combine into (5.4). □

Let

$$\theta_* = 0.3144831759741\ldots, \tag{5.5}$$

the exponent in the current Li–Yang bound $\Delta(x) = O_\varepsilon(x^{\theta_*+\varepsilon})$ for the Dirichlet divisor problem [8].

Proposition 5.3 (Extreme folded range)**.** *For every fixed $\varepsilon > 0$, uniformly for $0 < t \le 1/2$,*

$$M_p(t) \ll pt\log p + p^{\theta_*+\varepsilon}. \tag{5.6}$$

Consequently,

$$H_p(t) \ll pt\log p + p^{\theta_*+\varepsilon}, \tag{5.7}$$

and, for sufficiently small fixed $\varepsilon > 0$,

$$\int_0^{p^{-2/3}} |H_p(t)|\,|\cot(\pi t)|\,dt \ll p^{1/3}\log p. \tag{5.8}$$

Proof. For each counted denominator let

$$s_k = k\left\|\frac{p}{k}\right\| = \min(p \bmod k,\, k - (p \bmod k)).$$

Then s_k is a positive integer, $s_k \le pt$, and $k \mid p - s_k$ or $k \mid p + s_k$. Hence

$$M_p(t) \le \sum_{1 \le s \le pt} \big(d(p-s) + d(p+s)\big).$$

Apply the Li–Yang estimate for the summatory divisor function at the two endpoints of the interval $[p - pt, p + pt]$. This gives (5.6). Since $F_{\mathrm{fold}}(0) = 0$ and $F'_{\mathrm{fold}}(t) = \rho(t) + \rho(1-t)$ is bounded on $[0, 1/2]$, (5.3) gives (5.7).

For $t < 1/p$, $M_p(t) = 0$, hence $H_p(t) = O(pt)$. On $[1/p, p^{-2/3}]$, use (5.7) and $|\cot(\pi t)| \ll 1/t$. The $pt \log p$ term integrates to $O(p^{1/3} \log p)$, while the divisor-error term contributes $O(p^{\theta_*+\varepsilon} \log p) = o(p^{1/3} \log p)$ if $\varepsilon < 1/3 - \theta_*$. □

Corollary 5.4 (Mesoscopic location of the remaining logarithm)**.** *Theorem 3.5 can be recovered from (5.4), and the only range in that argument capable of producing the second logarithm is*

$$p^{-2/3} \le t \le \frac{1}{2}. \tag{5.9}$$

Indeed,

$$\int_{p^{-2/3}}^{1/2} |H_p(t)|\,|\cot(\pi t)|\,dt \ll p^{1/3}(\log p)^2 \tag{5.10}$$

by $\|H_p\|_\infty \ll p^{1/3} \log p$.

5.1. **Exact regular-grid representation.** To remove endpoint ambiguity, define the strict count

$$N_p^<(t) = \#\left\{2 \le k \le p-1 : \left\{\frac{p}{k}\right\} < t\right\}, \quad D_p^<(t) = N_p^<(t) - pF(t) + 2t, \tag{5.11}$$

and $H_p^<(t) = D_p^<(t) - D_p^<(1-t)$.

Theorem 5.5 (Exact reciprocal sawtooth identity)**.** *For* $0 < t < 1$,

$$\begin{aligned} H_p^<(t) = &\sum_{m=1}^{p} \left(\left\{\frac{p}{m+t}\right\} - \left\{\frac{p}{m+1-t}\right\}\right) \\ &+ p\left[\psi_0(p+2-t) - \psi_0(p+1+t)\right] + 4t - 2. \end{aligned} \tag{5.12}$$

At a jump, switching from $<$ to $\le$ adds exactly the corresponding jump multiplicity. Away from jumps, $H_p^< = H_p$.

Proof. Include temporarily the two endpoint denominators $k = 1, p$. For strict counting, partition by $m = \lfloor p/k \rfloor$:

$$\#\left\{1 \le k \le p : \left\{\frac{p}{k}\right\} < t\right\} = \sum_{m=1}^{p} \left(\left\lfloor \frac{p}{m} \right\rfloor - \left\lfloor \frac{p}{m+t} \right\rfloor\right).$$

Use $\lfloor x \rfloor = x - \{x\}$ and

$$\sum_{m=1}^{p} \left(\frac{1}{m} - \frac{1}{m+t}\right) = \psi_0(p+1) - \psi_0(p+1+t) + F(t).$$

After removing $k = 1, p$, this yields an exact formula for $D_p^<(t)$. Subtract the same formula with t replaced by $1-t$: the terms $\{p/m\}$ and $\psi_0(p+1)$ cancel, leaving (5.12). □

The smooth correction in (5.12) satisfies

$$p\left[\psi_0(p+2-t) - \psi_0(p+1+t)\right] + 4t - 2 = 2t - 1 + O(p^{-1}) \tag{5.13}$$

uniformly for $0 \le t \le 1$. This explains why omitting it in numerical reconstructions produces a bounded, t-dependent residual rather than a divergent error.

Remark 5.6 (The reciprocal variable is essential). The identity (5.12) uses the reciprocal quantities $p/(m+t)$ and $p/(m+1-t)$, not the linear quantities $(m+t)/p$ and $(m+1-t)/p$. The linear substitution carries no arithmetic content: for $0 < t < 1/2$,

$$\sum_{m=1}^{p} \left(\left\{\frac{m+t}{p}\right\} - \left\{\frac{m+1-t}{p}\right\}\right) = 2t - 1 \tag{5.14}$$

exactly, for every integer $p \geq 1$, prime or not. Indeed, for $1 \leq m \leq p-1$ both arguments already lie in $(0,1)$, so the braces are inert and each term equals $(2t-1)/p$; the term $m = p$ contributes the same value directly. Summing the p equal terms gives (5.14). In particular, a claimed identity of the form $H_p(t) = \sum_{m=1}^{p}(\{(m+t)/p\} - \{(m+1-t)/p\}) + O(1)$ is equivalent to $H_p(t) = O(1)$ uniformly in p, which contradicts Remark 5.8 and the exhaustive computations of Section 12. The reciprocal form $p/(m+t)$ is not an equivalent rewriting: its derivative in m is $-p/(m+t)^2$, so it crosses $\asymp p/m^2$ integers per unit step and carries exactly the hyperbolic geometry that Theorem 3.4 is built to control. This is the reason (5.12) is stated with $p/(m+t)$.

Lemma 5.7 (Where the continuous maximum occurs)**.** *On every open interval between consecutive jump points of M_p, H_p is strictly decreasing. Hence the supremum of $|H_p|$ on $[0,1/2]$ is attained as a one-sided value at a jump.*

Proof. Between jumps, (5.3) gives

$$H_p'(t) = 4 - p(\rho(t) + \rho(1-t)).$$

The function in parentheses is convex and symmetric about $1/2$, hence its minimum is

$$2\psi_1(3/2) = \pi^2 - 8 > 4/3.$$

Thus $H_p'(t) < 0$ for every prime $p \geq 3$. □

Remark 5.8 (Numerical conjecture). The exhaustive jump computations in Section 12 suggest

$$\sup_{p^{-2/3} \leq t \leq 1/2} |H_p(t)| \ll p^{1/3}. \tag{5.15}$$

No proof of (5.15) is given. If it holds, then Theorem 5.2 and Proposition 5.3 imply the sharper product bound

$$\log|P(p)| = -Cp + O(p^{1/3}\log p). \tag{5.16}$$

6. Integer singular core and complementary-divisor factorization

The logarithmic singularity can be separated from a smooth bounded-variation term. For a prime p, set

$$s_k = k\left\|\frac{p}{k}\right\| = \min(p \bmod k, k - (p \bmod k)), \qquad 2 \leq k \leq p-1, \tag{6.1}$$

and

$$h(u) = \log\frac{\sin \pi u}{\pi u}, \qquad 0 < u \leq \frac{1}{2}, \quad h(0) = 0. \tag{6.2}$$

Then $h'(u) = \pi\cot(\pi u) - 1/u < 0$ and

$$\operatorname{Var}_{[0,1/2]} h = \log\frac{\pi}{2}. \tag{6.3}$$

Proposition 6.1 (Exact smooth/integer split)**.** *Let*

$$S_p = \prod_{k=2}^{p-1} s_k. \tag{6.4}$$

Then

$$\log|P(p)| = (p-2)\log\pi + \log S_p - \log((p-1)!) + \sum_{k=2}^{p-1} h\left(\frac{s_k}{k}\right). \tag{6.5}$$

Define

$$J_0 = \int_0^{1/2} \log u \, [\rho(u) + \rho(1-u)] \, du, \tag{6.6}$$

$$H_0 = \int_0^{1/2} h(u) \, [\rho(u) + \rho(1-u)] \, du. \tag{6.7}$$

Then

$$\log \pi + J_0 + H_0 = -C, \tag{6.8}$$

and, with

$$\mathcal{A}(p) = \log S_p - \log((p-1)!) - pJ_0, \tag{6.9}$$

we have

$$\log |P(p)| + Cp = \mathcal{A}(p) - 2\log \pi + O(p^{1/3} \log p). \tag{6.10}$$

Proof. The identity

$$\log \sin(\pi u) = \log \pi + \log u + h(u)$$

with $u = s_k/k$ gives (6.5). Folding the density gives (6.8). Since h has bounded variation independent of p, Theorem 3.4 applied to the folded distribution gives

$$\sum_{k=2}^{p-1} h(s_k/k) = pH_0 + O(p^{1/3} \log p).$$

Substitution proves (6.10). □

Numerically,

$$J_0 = -1.750632969245044\ldots, \qquad H_0 = -0.137891778443756\ldots. \tag{6.11}$$

Thus every fluctuation larger than $p^{1/3} \log p$ in the proved $O(p^{1/3}(\log p)^2)$ remainder must reside in the explicit integer core $\mathcal{A}(p)$.

6.1. **A symmetric shifted-divisor count.** For an integer S with $1 \le S < (p-1)/2$, let

$$\mathcal{N}_p(S) = \#\{2 \le k \le p-1 : s_k \le S\}. \tag{6.12}$$

Proposition 6.2. *One has the exact identity*

$$\begin{aligned} \mathcal{N}_p(S) = S - 2 + \sum_{n=p-S}^{p+S} d(n) \\ - \sum_{k=1}^{2S} \left(\left\lfloor \frac{p+S}{k} \right\rfloor - \left\lfloor \frac{p-S-1}{k} \right\rfloor \right). \end{aligned} \tag{6.13}$$

Proof. All $2 \le k \le 2S$ satisfy $s_k \le k/2 \le S$. For $k > 2S$, the condition $s_k \le S$ is equivalent to the existence of a unique integer q such that $|p - qk| \le S$. Counting the pairs (q,k) by the integer $n = qk \in [p-S, p+S]$, and then subtracting the divisors $k \le 2S$, gives (6.13); the divisor $k = 1$ must be included in the correction sum. □

6.2. **Complementary divisors and the corrected gamma factorization.** Every $k \geq 3$ belongs to exactly one of the branches $p - s_k = qk$ or $p + s_k = qk$. The condition $k > 2s_k$ gives

$$\begin{aligned} \log S_p = &\sum_{q\geq 1} \sum_{\substack{s\equiv p \ (q) \\ (2q+1)s<p}} \log s \\ &+ \sum_{q\geq 2} \sum_{\substack{s\equiv -p \ (q) \\ (2q-1)s<p}} \log s. \end{aligned} \tag{6.14}$$

The exceptional denominator $k = 2$ has $s_2 = 1$ and does not affect the logarithm.

For $q \geq 2$, put

$$a_q = p \bmod q, \quad \alpha_q = \frac{a_q}{q}, \quad X_q^- = \left\lfloor \frac{p-1}{2q+1} \right\rfloor, \quad X_q^+ = \left\lfloor \frac{p-1}{2q-1} \right\rfloor, \tag{6.15}$$

and

$$n_-(q) = \max\left(0, 1 + \left\lfloor \frac{X_q^- - a_q}{q} \right\rfloor\right), \tag{6.16}$$

$$n_+(q) = \max\left(0, 1 + \left\lfloor \frac{X_q^+ - (q - a_q)}{q} \right\rfloor\right). \tag{6.17}$$

Proposition 6.3 (Corrected gamma factorization)**.** *For $q \geq 2$, define*

$$\begin{aligned} \mathcal{G}_q(p) = q^{n_-(q)+n_+(q)} &\frac{|\sin(\pi p/q)|}{\pi} \\ &\times \Gamma(n_-(q) + \alpha_q)\, \Gamma(n_+(q) + 1 - \alpha_q)\,. \end{aligned} \tag{6.18}$$

Then

$$S_p = \left\lfloor \frac{p-1}{3} \right\rfloor! \prod_{q=2}^{(p-1)/2} \mathcal{G}_q(p). \tag{6.19}$$

Proof. The first progression in (6.14) is $a_q, a_q + q, \ldots, a_q + (n_- - 1)q$, whose product is

$$q^{n_-} \frac{\Gamma(n_- + \alpha_q)}{\Gamma(\alpha_q)}.$$

The second progression starts at $q - a_q$ and contributes

$$q^{n_+} \frac{\Gamma(n_+ + 1 - \alpha_q)}{\Gamma(1 - \alpha_q)}.$$

Euler reflection gives

$$\frac{1}{\Gamma(\alpha_q)\Gamma(1-\alpha_q)} = \frac{\sin(\pi\alpha_q)}{\pi} = \frac{|\sin(\pi p/q)|}{\pi}.$$

For $q = 1$, the first branch contributes $1 \cdot 2 \cdots \lfloor (p-1)/3 \rfloor$. Multiplication gives (6.19). □

Proposition 6.4 (Uniform low-complementary-divisor smoothing)**.** *Let $2 \leq Q \leq p^{1/3}$. Put*

$$Y_q^- = \frac{p-1}{q(2q+1)}, \qquad Y_q^+ = \frac{p-1}{q(2q-1)}$$

and

$$\begin{aligned} \mathcal{M}_q(p) = &\left(Y_q^- - \frac{1}{2}\right) \log \frac{p-1}{2q+1} - Y_q^- \\ &+ \left(Y_q^+ - \frac{1}{2}\right) \log \frac{p-1}{2q-1} - Y_q^+. \end{aligned} \tag{6.20}$$

Then

$$\sum_{q=2}^{Q}\log \mathcal{G}_q(p)=\sum_{q=2}^{Q}\mathcal{M}_q(p)+O\left(Q\log p+\frac{Q^3}{p}\right). \tag{6.21}$$

In particular, at $Q=\lfloor p^{1/3}\rfloor$,

$$\sum_{q\le p^{1/3}}\log \mathcal{G}_q(p)=\sum_{q\le p^{1/3}}\mathcal{M}_q(p)+O(p^{1/3}\log p). \tag{6.22}$$

Proof. For $q\le Q$, the two large gamma arguments in (6.18) have the form

$$z_q^-=Y_q^-+\eta_q^-,\qquad z_q^+=Y_q^++\eta_q^+,\qquad 0<\eta_q^{\pm}\le 1,$$

with $Y_q^{\pm}\asymp p/q^2$. Stirling's formula, uniformly in this range, gives

$$\log\Gamma(z)=\left(z-\frac12\right)\log z-z+\frac12\log(2\pi)+O(z^{-1}).$$

Moving $z_q^{\pm}$ to $Y_q^{\pm}$ costs $O(\log p)$ per q, while the two Stirling remainders sum to $O(\sum_{q\le Q}q^2/p)=O(Q^3/p)$. Finally $|\sin(\pi p/q)|\gg 1/q$, so the explicit sine factor costs only $O(\log q)$ per q, already contained in $O(Q\log p)$. □

Remark 6.5. The gamma factorization is exact and the low-q smoothing is a genuine localization statement. It does not, by itself, improve Theorem 3.5: substituting the reflected sine factors back into the complete product reconstructs a complementary part of the original product. Any sharper global estimate must therefore control the remaining high-q shifted-divisor sector or the folded discrepancy, rather than rely on gamma reflection alone.

7. Primitive divisor decomposition

To obtain finite logarithms for every integer, omit precisely the zero divisor factors:

$$\widetilde{P}(n)=\prod_{\substack{2\le k\le n-1\\ k\nmid n}}\left|\sin\left(\frac{\pi n}{k}\right)\right|,\qquad \widetilde{\lambda}(n)=\log\widetilde{P}(n). \tag{7.1}$$

Define $L(1)=0$ and, for $d\ge 2$,

$$L(d)=\sum_{\substack{2\le k\le d-1\\ (k,d)=1}}\log\sin\left(\frac{\pi(d \bmod k)}{k}\right). \tag{7.2}$$

Theorem 7.1. *For every* $n\ge 1$,

$$\widetilde{\lambda}(n)=\sum_{d\mid n}L(d),\qquad L=\mu*\widetilde{\lambda}. \tag{7.3}$$

Proof. For $k\nmid n$, put $g=(n,k)$, $d=n/g$, and $k=gk'$. Then $(d,k')=1$, $2\le k'<d$, and $\{n/k\}=\{d/k'\}$. Conversely, a divisor $d\mid n$ and a primitive index $2\le k'<d$ reconstruct $k=(n/d)k'$. This is a bijection between the factors on the two sides. The second formula is the standard Möbius inversion formula [1, Chapter 2]. □

The same bijection works for every function of $\{n/k\}$; thus the divisor convolution itself is not sine-specific. With

$$R_L(d)=L(d)+C\varphi(d),\qquad \widetilde{R}(n)=\widetilde{\lambda}(n)+Cn, \tag{7.4}$$

the classical identity $\sum_{d\mid n}\varphi(d)=n$ gives

$$\widetilde{R}=\mathbf{1}*R_L. \tag{7.5}$$

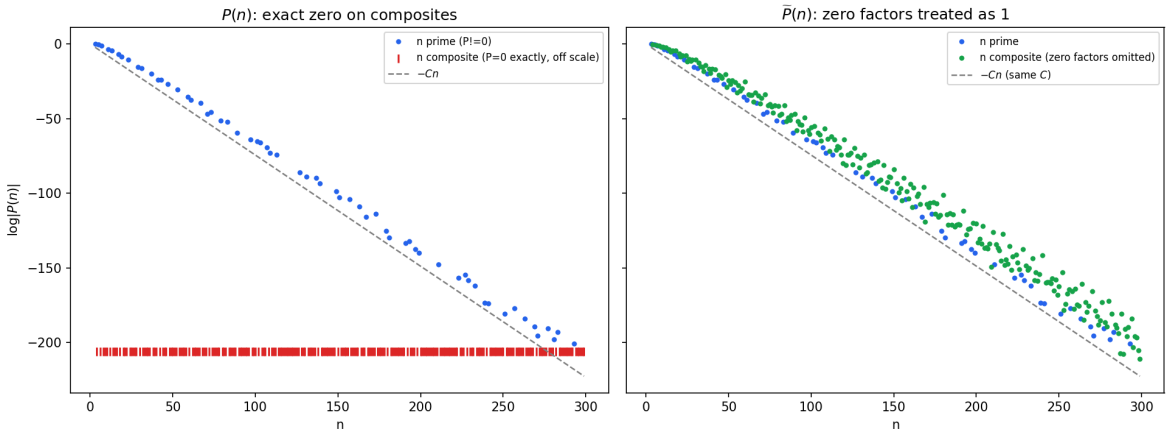


FIGURE 4. $P(n)$ versus $\widetilde{P}(n)$, $3 \le n \le 300$, on a logarithmic scale. Left: $P(n)$ is exactly zero on composites, so only primes appear. Right: omitting the zero divisor factors restores every integer to the same line $-Cn$, with the identical constant C of Theorem 3.2 governing primes and composites alike. This is the content of Corollary 7.2.

Corollary 7.2. *Uniformly for $n \ge 2$,*

$$\widetilde{\lambda}(n) = -Cn + O\Big(n^{1/3}(\log n)^2\Big), \qquad R_L(n) = O_\varepsilon(n^{1/3+\varepsilon}). \tag{7.6}$$

Proof. Apply Theorem 3.4 to the full reciprocal fractional-part multiset and delete the zero factors $k \mid n$. This changes the distribution function by at most $d(n)$, and every remaining logarithm has size $O(\log n)$. Thus the proof of Theorem 3.5 gives $O(n^{1/3}(\log n)^2 + d(n)\log n)$, and the divisor term is absorbed. Möbius inversion in (7.5) gives

$$|R_L(n)| \le \sum_{d\mid n} |\widetilde{R}(n/d)| \ll n^{1/3}(\log n)^2 \sum_{d\mid n} d^{-1/3} = O_\varepsilon(n^{1/3+\varepsilon}).$$

□

For $\Re s > 4/3$, absolute convergence and (7.5) give the standard Dirichlet-series identity

$$A(s) = \zeta(s)B(s), \qquad A(s) = \sum_{n\ge1} \frac{\widetilde{R}(n)}{n^s}, \quad B(s) = \sum_{n\ge1} \frac{R_L(n)}{n^s}. \tag{7.7}$$

This is exactly the Dirichlet-series image of a divisor convolution; no continuation of A or B is asserted; compare [1, Chapter 2].

8. Euclidean Lambert expansion and centered kernel

Put

$$\ell(x) = \log\sin(\pi x), \qquad H_k(z) = \sum_{\substack{1\le r<k\\(r,k)=1}} \ell\Big(\frac{r}{k}\Big)\, z^r. \tag{8.1}$$

Theorem 8.1. *For $|z| < 1$,*

$$\mathcal{G}(z) := \sum_{n\ge2} L(n) z^n = \sum_{k\ge2} \frac{z^k}{1-z^k} H_k(z). \tag{8.2}$$

Moreover, $H_k(z) = z^k H_k(z^{-1})$ as an identity of Laurent polynomials.

Proof. Write $n = mk + r$ in every summand of $L(n)$, with $m \geq 1$, $1 \leq r < k$, and $(r,k) = 1$. Absolute convergence for $|z| < 1$ allows the rearrangement

$$\sum_{n\geq 2} L(n) z^n = \sum_{k\geq 2} \sum_{\substack{1\leq r<k \\ (r,k)=1}} \ell(r/k) z^r \sum_{m\geq 1} z^{mk}.$$

The involution $r \mapsto k - r$ proves the reflection identity. □

Theorem 8.2. *For $u \in \mathbb{R}$, define*

$$V_k(u) = e^{iku/2} H_k(e^{-iu}) - H_k(1). \tag{8.3}$$

Then V_k is real, even, and nonnegative, with

$$V_k(u) = 2 \sum_{\substack{1\leq r<k/2 \\ (r,k)=1}} \left[-\ell\left(\frac{r}{k}\right)\right] \left[1 - \cos\left(\left(\frac{k}{2} - r\right) u\right)\right]. \tag{8.4}$$

The kernel $V_k(u - v)$ is conditionally negative definite. Thus, if $\sum_j c_j = 0$,

$$\sum_{i,j} c_i \overline{c_j} V_k(u_i - u_j) \leq 0. \tag{8.5}$$

On the Laplace axis,

$$\Delta_k(t) = e^{kt/2} H_k(e^{-t}) - H_k(1) \leq 0 \qquad (t \geq 0). \tag{8.6}$$

Moreover, for every $j \geq 1$,

$$\Delta_k^{(2j)}(0) = 2 \sum_{\substack{1\leq r<k/2 \\ (r,k)=1}} \ell(r/k) \left(\frac{k}{2} - r\right)^{2j} \leq 0, \tag{8.7}$$

with strict inequality whenever the reduced-residue set contains a point distinct from $k/2$.

Proof. Pair r with $k - r$. Since $\ell(r/k) < 0$ away from the central point, the pairing gives (8.4). Writing $a_{k,r} = k/2 - r$ and $w_{k,r} = -2\ell(r/k) > 0$, the quadratic form in (8.5) equals

$$-\frac{1}{2} \sum_r w_{k,r} \left(\left| \sum_j c_j e^{i a_{k,r} u_j} \right|^2 + \left| \sum_j c_j e^{-i a_{k,r} u_j} \right|^2 \right),$$

because the constant term is zero. This is the standard negative-type form of Schoenberg [13]. Replacing u by $-it$ gives

$$\Delta_k(t) = 2 \sum_r \ell(r/k) (\cosh(a_{k,r} t) - 1) \leq 0.$$

Differentiating this finite sum at the origin gives (8.7). □

Remark 8.3 (Universality of the centered sign). The proof of Theorem 8.2 uses only reflection symmetry under $r \mapsto k - r$ and the inequality $\ell(r/k) < 0$ away from the central point. Consequently, (8.4), conditional negative definiteness, (8.6), and (8.7) remain valid verbatim if the reduced residues are replaced by any subset $S_k \subseteq \{1, \ldots, k-1\}$ satisfying $r \in S_k \iff k - r \in S_k$. Coprimality therefore plays no role in these signs.

For the reduced-residue block, the arithmetic content is instead carried by the positive spectral measure

$$\nu_k = \sum_{\substack{1\leq r<k/2 \\ (r,k)=1}} \left[-\ell\left(\frac{r}{k}\right)\right] \left(\delta_{k/2-r} + \delta_{-(k/2-r)}\right), \tag{8.8}$$

for which

$$V_k(u) = \int_{\mathbb{R}} (1 - \cos(\xi u))\, d\nu_k(\xi).$$

Thus the atoms $k/2 - r$ and their weights, not positivity by itself, retain the reduced-residue arithmetic. In particular, the signs $\mathcal{D}(t) < 0$ and $\kappa > 0$ below are termwise signs; the asymptotic size and spectral distribution are the nontrivial information.

The complete value of a block is classical:

$$H_k(1) = \Lambda(k) - \varphi(k) \log 2. \tag{8.9}$$

Indeed, this is $\Phi_k(1) = e^{\Lambda(k)}$, discussed further in Section 10. The centered kernel subtracts this total mass before retaining the frequencies $k/2 - r$.

Define

$$\mathcal{D}(t) = \sum_{k \geq 2} \frac{e^{-3kt/2}}{1 - e^{-kt}} \Delta_k(t), \qquad t > 0. \tag{8.10}$$

Proposition 8.4. *As $t \downarrow 0$,*

$$\mathcal{D}(t) \sim -\frac{\kappa}{t^2}, \tag{8.11}$$

where

$$\kappa = -\frac{6}{\pi^2} \int_0^\infty \frac{y e^{-3y/2}}{1 - e^{-y}} \int_0^1 \ell(x) \left[\cosh\left(y \left(x - \frac{1}{2} \right) \right) - 1 \right] \mathrm{d}x\, \mathrm{d}y > 0. \tag{8.12}$$

Proof. Insert (8.6) into (8.10). With

$$F(y, x) = \frac{e^{-3y/2}}{1 - e^{-y}} \ell(x) (e^{-y(x-1/2)} - 1),$$

one obtains

$$t^2 \mathcal{D}(t) = t^2 \sum_{k \geq 2} \sum_{\substack{1 \leq r < k \\ (r,k)=1}} F(tk, r/k).$$

For fixed $0 < \delta < 1/2$ and $0 < \eta < M < \infty$, truncate to $\delta \leq r/k \leq 1 - \delta$ and $\eta \leq tk \leq M$. On this compact region the weight is bounded and continuous, so the classical primitive-lattice Riemann-sum limit obtained by Möbius inversion gives

$$t^2 \sum_{k \geq 1} \sum_{\substack{1 \leq r < k \\ (r,k)=1}} F(tk, r/k) \longrightarrow \frac{6}{\pi^2} \int_0^\infty y \int_0^1 F(y, x)\, \mathrm{d}x\, \mathrm{d}y$$

after the three truncations are removed; see [5, Chapter XVIII]. We now justify that removal.

For $0 < x \leq 1/2$,

$$|\ell(x)| \ll 1 + \log(1/x), \tag{8.13}$$

and the same bound holds near 1 by reflection. Also

$$\left| \frac{e^{-3y/2}}{1 - e^{-y}} (e^{-y(x-1/2)} - 1) \right| \leq \frac{y e^{-y}}{2(1 - e^{-y})} =: h(y),$$

where $h(y) = O(1)$ as $y \downarrow 0$ and $h(y) \ll y e^{-y}$ as $y \to \infty$. From (8.13) and Stirling's estimate for $\sum_{r \leq \delta k} \log r$, uniformly in k,

$$\sum_{1 \leq r \leq \delta k} |\ell(r/k)| \ll k\delta(1 + |\log \delta|) + \log(2k). \tag{8.14}$$

Hence the two endpoint strips contribute, in the limit superior as $t \downarrow 0$, at most $O(\delta(1 + |\log \delta|))$; the term containing $\log(2k)$ is $o(1)$.

For the y-tails, the complete sine product gives the exact bound

$$\sum_{r=1}^{k-1} |\ell(r/k)| = (k-1)\log 2 - \log k \ll k.$$

Therefore the range $tk < \eta$ contributes $O(\eta^2)$, while the range $tk > M$ contributes

$$O\Big(\int_M^\infty yh(y)\,\mathrm{d}y\Big) = O\Big((1+M)^2 e^{-M}\Big),$$

uniformly for sufficiently small t. Let first $t \downarrow 0$, then $\delta, \eta \downarrow 0$, and finally $M \to \infty$. This proves the untruncated limit despite the logarithmic endpoint singularities.

Finally, the symmetry $\ell(x) = \ell(1-x)$ replaces the exponential bracket by the hyperbolic cosine in (8.12). The resulting integral is absolutely convergent by the same bounds. Since $\ell(x) < 0$ for $0 < x < 1$, except at $x = 1/2$, and $\cosh(y(x-1/2)) - 1 \geq 0$, the inner integral in (8.12) is strictly negative for every $y > 0$; hence $\kappa > 0$. □

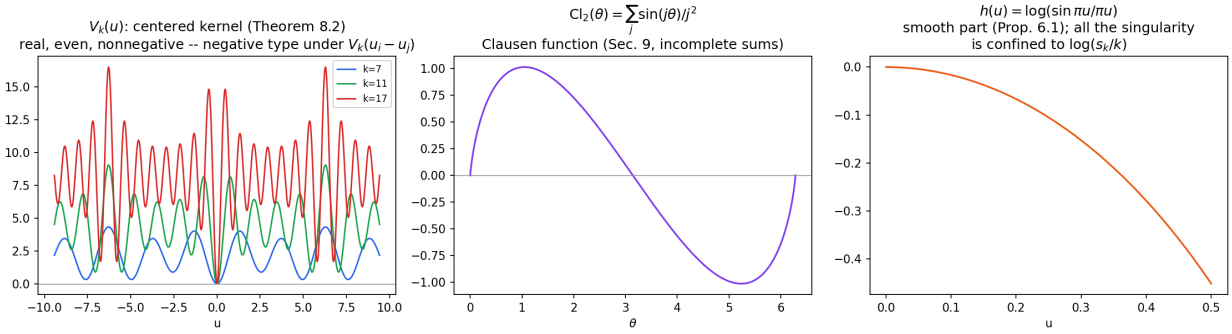


FIGURE 5. Three further objects used in this paper, none of which coincide with ρ, ℓ, K_J, D_p, or H_p of Figure 6. Left: the centered kernel $V_k(u)$ of Theorem 8.2, shown for $k = 7, 11, 17$; it is real, even, and nonnegative, as proved there. Center: the Clausen function $\mathrm{Cl}_2(\theta)$ of Section 9. Right: the bounded-variation kernel $h(u)$ of (6.2), which carries no singularity, in contrast with $\ell(x) = \log\sin(\pi x)$; all of the logarithmic singularity of the product is confined instead to the integer core $\log(s_k/k)$ of Proposition 6.1.

9. Incomplete reduced-residue sums

For $0 \leq S < m$, put

$$\mathcal{H}(m,S) = \sum_{s=1}^{S} \log\Big(2\sin\frac{\pi s}{m}\Big), \qquad \mathrm{Cl}_2(\theta) = \sum_{j\geq 1} \frac{\sin(j\theta)}{j^2}. \tag{9.1}$$

The Clausen function and its relation to the dilogarithm are classical; see [7].

Proposition 9.1. *For $q > 1$ and $0 \leq R < q$,*

$$\sum_{\substack{1\leq r\leq R\\(r,q)=1}} \log\Big(2\sin\frac{\pi r}{q}\Big) = \sum_{e\mid q} \mu(e)\,\mathcal{H}\Big(\frac{q}{e}, \Big\lfloor\frac{R}{e}\Big\rfloor\Big). \tag{9.2}$$

Proof. Insert $\mathbf{1}_{(r,q)=1} = \sum_{e\mid(r,q)} \mu(e)$ and write $r = es$. Then $r \leq R$ is equivalent to $s \leq \lfloor R/e \rfloor$, while $r/q = s/(q/e)$. □

The floor in (9.2) is essential:

$$\frac{\lfloor R/e \rfloor}{q/e} = \frac{R - (R \bmod e)}{q}. \tag{9.3}$$

Thus the Clausen phases are not exactly independent of e.

Lemma 9.2. *For $1 \le S < m$, with $a = S/m$,*

$$\mathcal{H}(m,S) = -\frac{m}{2\pi}\,\mathrm{Cl}_2(2\pi a) + \frac{1}{2}\log(2\pi S) + \frac{1}{2}\log\!\left(\frac{\sin(\pi a)}{\pi a}\right) + E(m,S), \tag{9.4}$$

$$|E(m,S)| \le \frac{1}{12S} + \frac{1}{2}\log\!\left(\frac{\pi a}{\sin(\pi a)}\right). \tag{9.5}$$

Proof. Split

$$\log(2\sin\pi x) = \log(2\pi x) + g(x), \qquad g(x) = \log\!\left(\frac{\sin\pi x}{\pi x}\right).$$

Stirling's formula with its elementary remainder treats the first term. For the second, first-order Euler summation gives

$$\sum_{s=1}^{S} g(s/m) = m\int_0^a g(x)\,\mathrm{d}x + \frac{1}{2}g(a) + E_g, \qquad |E_g| \le \frac{1}{2}\int_0^a |g'(x)|\,\mathrm{d}x = -\frac{1}{2}g(a).$$

Finally,

$$\int_0^S \log\!\left(2\sin\frac{\pi t}{m}\right)\mathrm{d}t = -\frac{m}{2\pi}\,\mathrm{Cl}_2(2\pi S/m).$$

These are the standard Euler–Maclaurin ingredients for partial rational sine products; compare [10]. □

Theorem 9.3. *For $q > 1$ and $0 \le R < q$,*

$$\sum_{\substack{1\le r\le R\\(r,q)=1}} \log\!\left(2\sin\frac{\pi r}{q}\right) = -\frac{\varphi(q)}{2\pi}\,\mathrm{Cl}_2\!\left(2\pi\frac{R}{q}\right) + \frac{1}{2}\Lambda(q) + \varepsilon(q,R), \tag{9.6}$$

where, uniformly in R,

$$|\varepsilon(q,R)| \ll d(q)\log(2q). \tag{9.7}$$

Moreover,

$$\sum_{R=0}^{q-1} \varepsilon(q,R) = 0. \tag{9.8}$$

Proof. For $R = 0$, take $\varepsilon(q,0) = -\Lambda(q)/2$. Suppose $R \ge 1$. Apply Lemma 9.2 to every divisor $e \mid q$ with $e \le R$, and put $a_e = e\lfloor R/e \rfloor/q$ and $a = R/q$. Since $|a - a_e| < e/q$, the classical modulus estimate

$$|\,\mathrm{Cl}_2(2\pi x) - \mathrm{Cl}_2(2\pi y)| \ll |x-y|\log\frac{2}{|x-y|}$$

follows by integrating $\mathrm{Cl}_2'(\theta) = -\log|2\sin(\theta/2)|$. It shows that replacing a_e by a, after multiplication by q/e, costs $O(1+\log(q/e))$ per divisor. The divisors $e > R$ satisfy the same total bound because $|\,\mathrm{Cl}_2(2\pi a)| \ll a\log(2/a)$. The Clausen terms then collapse by $\sum_{e|q}\mu(e)/e = \varphi(q)/q$.

For $e \le R$, $\log\lfloor R/e\rfloor = \log R - \log e + O(1)$. Completing the divisor sums and using the known identities

$$\sum_{e|q}\mu(e) = 0, \qquad \sum_{e|q}\mu(e)\log e = -\Lambda(q)$$

extracts $\Lambda(q)/2$. The sinc terms and the remainders in (9.5) contribute at most $O(d(q)\log(2q))$. This proves (9.7).

For the mean, pair r with $q-r$ and use the complete cyclotomic product to obtain

$$\sum_{R=0}^{q-1}\sum_{\substack{r\le R\\(r,q)=1}}\log\Big(2\sin\frac{\pi r}{q}\Big)=\frac{q}{2}\Lambda(q).$$

Also $\sum_{R=0}^{q-1}\mathrm{Cl}_2(2\pi R/q)=0$. Summing (9.6) over R gives (9.8). □

At the moving cutoffs relevant to the diagonal product, define

$$\mathcal{E}(X)=\sum_{q=2}^{X-1}\varepsilon(q,X \bmod q). \tag{9.9}$$

The uniform bound (9.7) and the elementary estimate $\sum_{q\le X}d(q)\ll X\log X$ give only

$$\mathcal{E}(X)\ll X(\log X)^2. \tag{9.10}$$

Thus the pointwise Clausen decomposition does not, by direct summation of its remainders, reach the $\sqrt{X}$-scale; any sharper use of the formula must exploit cancellation or additional structure in the moving remainders.

The totient-weighted Clausen aggregate has an elementary quadratic mean.

Proposition 9.4. *As $X\to\infty$ through the integers,*

$$\sum_{q<X}\varphi(q)\,\mathrm{Cl}_2\Big(2\pi\Big\{\frac{X}{q}\Big\}\Big)=c_{\mathrm{Cl}}X^2+o(X^2), \tag{9.11}$$

where

$$c_{\mathrm{Cl}}=\frac{6}{\pi^2}\int_0^1\mathrm{Cl}_2(2\pi t)\sum_{m\ge1}\frac{\mathrm{d}t}{(m+t)^3}. \tag{9.12}$$

Proof. Set $f(t)=\mathrm{Cl}_2(2\pi t)$ and $g(u)=f(\{1/u\})$. Since $f(0)=f(1)=0$, the bounded function g is Riemann integrable on $[0,1]$. Partial summation with the classical estimate $\sum_{n\le y}\varphi(n)=3y^2/\pi^2+O(y\log y)$ gives

$$\frac{1}{X^2}\sum_{q<X}\varphi(q)g(q/X)\longrightarrow\frac{6}{\pi^2}\int_0^1 ug(u)\,\mathrm{d}u.$$

Substitute $v=1/u$ and split v into the intervals $[m,m+1]$ to obtain (9.12). The mean value of φ is standard; see [1, Chapter 3]. □

10. Arithmetic encodings

This section records finite codings obtained from the zeros and complete reduced-residue values of the product. The prime indicator, the Boolean Möbius formula, and the cyclotomic recovery of Λ are exact encodings; they are not analytic estimates. The exact Chebyshev reduction and the totient obstruction, which make genuinely different statements, are separated into the next section.

10.1. **Prime counting and a normalized count.** Adopt the empty-product convention $P(2)=1$. Since $P(n)=0$ exactly when n has a divisor k with $2\le k<n$,

$$\pi(X)=1+\sum_{n=3}^{X}\lceil|P(n)|\rceil\,. \tag{10.1}$$

This is an indicator formula of the same general kind as other exact but computationally inefficient prime formulas; compare Willans [16].

The exponential scale supplies a numerically stable normalization. Put

$$\beta_c = \frac{\log 2}{C}, \qquad S_c(X) = \sum_{2\le n\le X} |P(n)|^{\beta_c/n}, \tag{10.2}$$

where $0^a = 0$ for $a > 0$. For a prime p, write $R(p) = \log|P(p)| + Cp$. Theorem 3.5 gives $R(p) = O(p^{1/3}(\log p)^2)$, hence

$$|P(p)|^{\beta_c/p} = \frac{1}{2} + O\Big(p^{-2/3}(\log p)^2\Big). \tag{10.3}$$

The elementary Chebyshev bound $\pi(x) \ll x/\log x$ and partial summation imply

$$S_c(X) = \frac{1}{2}\pi(X) + O(X^{1/3}\log X). \tag{10.4}$$

Consequently, the classical von Koch criterion may still be written as

$$\text{RH} \iff 2S_c(X) - \mathrm{Li}(X) = O(\sqrt{X}\log X). \tag{10.5}$$

Indeed,

$$2S_c(X) - \mathrm{Li}(X) = \pi(X) - \mathrm{Li}(X) + O(X^{1/3}\log X), \tag{10.6}$$

and the added term is $o(\sqrt{X}\log X)$. Thus the strengthened product remainder sharpens the approximation of S_c to $\pi/2$, but it does not sharpen or alter the von Koch criterion itself; see [14, Chapter XIV].

10.2. **A Boolean formula for the Möbius function.** Define the Boolean quantities

$$A(k) = \lceil |P(k)| \rceil, \quad B_k(n) = \left\lceil \left| \sin \frac{\pi n}{k} \right| \right\rceil, \quad C_k(n) = \left\lceil \left| \sin \frac{\pi n}{k^2} \right| \right\rceil. \tag{10.7}$$

Proposition 10.1. *For every $n \ge 1$,*

$$\mu(n) = \prod_{k=2}^{n} \left[1 - A(k) + A(k)C_k(n)\big(2B_k(n) - 1\big)\right]. \tag{10.8}$$

Proof. If k is composite, then $A(k) = 0$ and the k-factor is 1. If k is prime, the factor is 1, -1, or 0 according as $k \nmid n$, $k \mid n$ but $k^2 \nmid n$, or $k^2 \mid n$. The product is therefore zero on nonsquarefree integers and otherwise is $(-1)^{\omega(n)}$, the standard definition of $\mu(n)$. □

Formula (10.8) is a finite Boolean encoding of prime exponents. It does not avoid the primality and divisibility tests encoded by its sine zeros.

10.3. **Cyclotomic recovery of the von Mangoldt function.** For $q \ge 2$, define the complete reduced-residue sum

$$J(q) = \sum_{\substack{1\le r<q\\(r,q)=1}} \log\sin\left(\frac{\pi r}{q}\right). \tag{10.9}$$

The classical cyclotomic evaluation

$$\prod_{\substack{1\le r<q\\(r,q)=1}} 2\sin\left(\frac{\pi r}{q}\right) = \Phi_q(1) = e^{\Lambda(q)} \tag{10.10}$$

gives the known identity

$$J(q) = \Lambda(q) - \varphi(q)\log 2. \tag{10.11}$$

See [6, Lemma 2.2] and [15, Chapters 1–2]. Thus the complete value $H_q(1)$ in (8.9) recovers $\Lambda(q)$; this is a cyclotomic identity, not a new formula for the distribution of prime powers.

11. Exact Chebyshev reduction and totient obstruction

The following results are not merely alternative codings of arithmetic functions. The first removes an auxiliary normalization exactly; the second shows that a natural termwise asymptotic decomposition is impossible at the scale relevant to the prime error term.

11.1. **Closed totient block and four-term identity.** Write $\Psi(X) = \sum_{m\le X}\Lambda(m)$. For an integer $X \ge 2$, let

$$T(X) = \sum_{n\le X}\varphi(n), \tag{11.1}$$

$$\mathcal{B}_X = \sum_{q=2}^{X-1}\left(\left\lfloor\frac{X}{q}\right\rfloor - 1\right)\varphi(q), \qquad Q_X = \sum_{q=2}^{X-1}\sum_{\substack{1\le r\le X \bmod q\\(r,q)=1}}\log\sin\left(\frac{\pi r}{q}\right). \tag{11.2}$$

Proposition 11.1 (Closed form for the totient block)**.** *For every integer $X \ge 2$,*

$$\mathcal{B}_X = \frac{X^2 - X}{2} + 1 - T(X). \tag{11.3}$$

Proof. The classical divisor identity $\sum_{d|m}\varphi(d) = m$ gives

$$\sum_{q\le X}\left\lfloor\frac{X}{q}\right\rfloor\varphi(q) = \sum_{m\le X}\sum_{q|m}\varphi(q) = \sum_{m\le X} m = \frac{X(X+1)}{2}.$$

Removing the terms $q = 1$ and $q = X$, and then subtracting $\sum_{q=2}^{X-1}\varphi(q) = T(X) - 1 - \varphi(X)$, yields (11.3) exactly. □

Proposition 11.2. *For every $X \ge 2$,*

$$\boxed{\Psi(X) = \log(X!) - (\log 2)\,\mathcal{B}_X + Q_X - \sum_{n\le X} L(n).} \tag{11.4}$$

In particular, the normalization constant C cancels identically.

Proof. Sum $L(n)$ first over $n \le X$ and then over its denominator q. For fixed q, the residues consist of $\lfloor X/q\rfloor - 1$ complete reduced-residue cycles and a terminal segment. Formula (10.11) gives

$$\sum_{n\le X} L(n) = \sum_{q=2}^{X-1}\left(\left\lfloor\frac{X}{q}\right\rfloor - 1\right)(\Lambda(q) - \varphi(q)\log 2) + Q_X.$$

The standard identity

$$\sum_{q\le X}\left\lfloor\frac{X}{q}\right\rfloor\Lambda(q) = \log(X!)$$

follows by summing $\log m = \sum_{q|m}\Lambda(q)$ over $m \le X$. Hence

$$\sum_{q=2}^{X-1}\left(\left\lfloor\frac{X}{q}\right\rfloor - 1\right)\Lambda(q) = \log(X!) - \Psi(X),$$

and rearrangement gives (11.4). Equivalently, in the normalization $R_L = L + C\varphi$, one has $\sum_{n\le X} R_L(n) = \sum_{n\le X} L(n) + CT(X)$, so the two $CT(X)$ terms in the previous normalized form cancel exactly. □

Combining Propositions 11.1 and 11.2 gives the four-term form

$$\Psi(X) = \log(X!) - (\log 2)\frac{X^2 - X + 2}{2} + \left[(\log 2)T(X) + Q_X - \sum_{n\le X} L(n)\right]. \tag{11.5}$$

This form makes an unavoidable cancellation visible.

11.2. **A proved obstruction to termwise estimation.**

Proposition 11.3 (Totient obstruction to termwise estimation)**.** *Let*

$$E_\varphi(X) = T(X) - \frac{3}{\pi^2}X^2. \tag{11.6}$$

Then

$$\limsup_{X\to\infty} \frac{|E_\varphi(X)|}{X} > 0. \tag{11.7}$$

Consequently, if

$$M_Q(X) = X - \log(X!) + (\log 2)\left(\frac{X^2 - X + 2}{2} - \frac{3}{\pi^2}X^2\right), \tag{11.8}$$

then

$$\limsup_{X\to\infty} \frac{1}{X}\left|Q_X - \sum_{n\le X} L(n) - M_Q(X)\right| > 0. \tag{11.9}$$

Thus the $\Omega(X)$ fluctuation in $Q_X - \sum_{n\le X} L(n)$ is forced by the totient term; it is larger than the $O(\sqrt{X}\log^2 X)$ precision required by the von Koch form of the Riemann-hypothesis error term for Ψ.

Proof. For a prime p,

$$\begin{aligned} E_\varphi(p) - E_\varphi(p-1) &= \varphi(p) - \frac{3}{\pi^2}(p^2 - (p-1)^2) \\ &= \left(1 - \frac{6}{\pi^2}\right)p + O(1). \end{aligned}$$

The coefficient is positive. Therefore at least one of $|E_\varphi(p)|$ and $|E_\varphi(p-1)|$ is at least a fixed positive multiple of p for every sufficiently large prime p, proving (11.7).

From (11.5),

$$Q_X - \sum_{n\le X} L(n) - M_Q(X) = \Psi(X) - X - (\log 2)E_\varphi(X). \tag{11.10}$$

The prime number theorem gives $\Psi(X) - X = o(X)$, whereas (11.7) supplies an $\Omega(X)$ term. This proves (11.9). In particular, replacing $T(X)$ by its quadratic main term before combining it with $Q_X - \sum_{n\le X} L(n)$ necessarily leaves an error of order X along a subsequence. The cancellation in the bracket of (11.5) must therefore be retained. □

Thus (11.4) remains a finite reorganization of the classical divisor identity for Λ, but (11.9) identifies precisely where a term-by-term asymptotic treatment fails: the dominant uncancelled fluctuation is totient rather than prime.

12. Numerical audit

Numerical experiment 12.1. All computations in this section check finite identities, constants, and finite-range conjectural diagnostics. None is used in a proof. The standalone script `v9_numeric_audit.py` reproduces the new tables, and `v9_figures.py` together with `v9_Px_funcion.py` regenerates Figures 1–5.

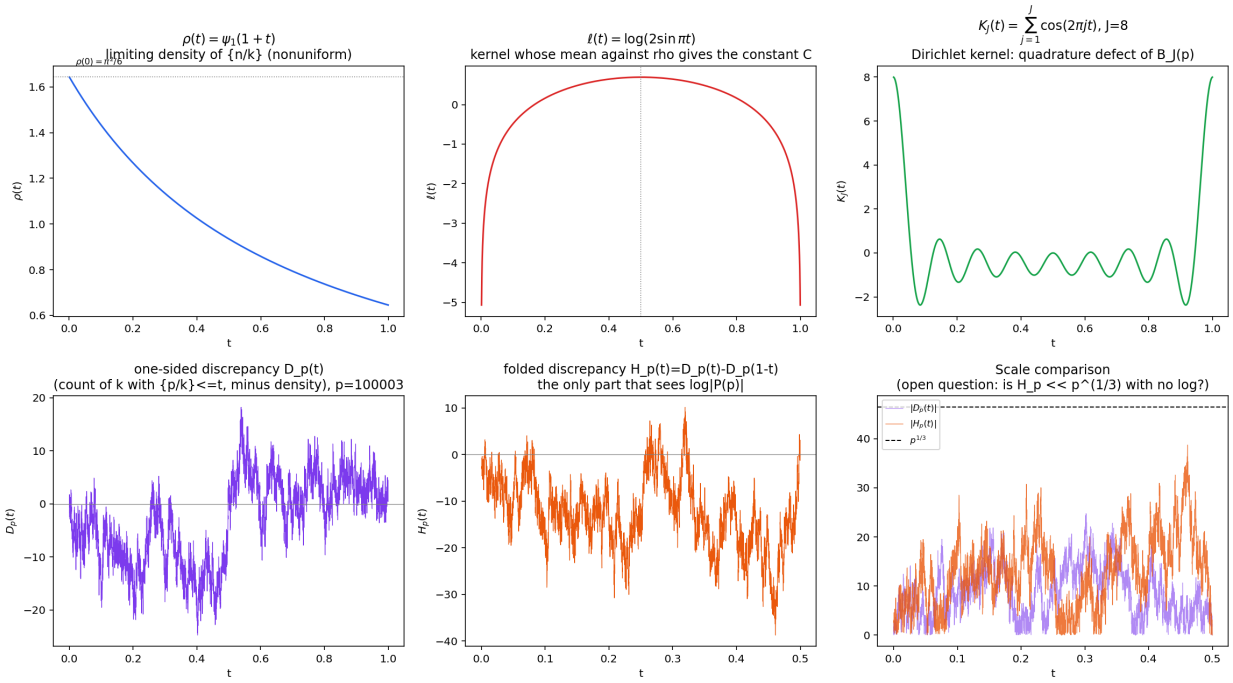


FIGURE 6. The central objects of Sections 3–5, all evaluated at the sample prime $p = 100003$ used throughout this section. Top row: the limiting density $\rho(t)$, the logarithmic sine kernel $\ell(t)$, and the Dirichlet kernel $K_J(t)$ for $J = 8$. Bottom row: the one-sided discrepancy $D_p(t)$, the folded discrepancy $H_p(t) = D_p(t) - D_p(1-t)$, and a direct comparison of $|D_p|$ and $|H_p|$ against $p^{1/3}$, illustrating Remark 5.8.

12.1. **Previously established finite checks.** Independent quadrature of (3.7) and summation of (3.10) agree to 7×10^{-13}, with

$$C = 0.7437948618394\ldots, \qquad C - \log 2 = 0.0506476813\ldots.$$

The divisor decomposition (7.3) was checked for every $n \le 1200$; the largest floating-point discrepancy was 2.3×10^{-13}.

For the exact cumulative-frequency object (4.8), direct evaluation at $p = 100003$ gives

$$\frac{B_{1000}}{\sqrt{1000p}} = -0.269309229, \quad \frac{B_{2000}}{\sqrt{2000p}} = -0.578604231, \quad \frac{B_{3000}}{\sqrt{3000p}} = -0.427901820.$$

These values are retained only as a diagnostic of the exact quadrature identity; Section 4 no longer uses a square-root B_J criterion as the best pointwise route.

The two sides of the Lambert identity (8.2), truncated at index 200, agreed to 8.7×10^{-19} at $z = 0.3$ and to 6.7×10^{-16} at $z = 0.7$. Direct quadrature gives $\kappa = 0.05826334\ldots$. The exact sieve (9.2) was checked for all $2 \le q \le 300$ and $0 \le R < q$, with maximum discrepancy 2.18×10^{-14}. The Boolean identity (10.8) and prime count (10.1) were checked through $n = 20000$; the closed totient form (11.3) has zero integer discrepancy for every $3 \le X \le 300$, and the simplified Chebyshev identity was checked at $X = 50, 100, 200, 300$ to 1.7×10^{-12}.

12.2. **Smooth/integer split and gamma audit.** Quadrature of (6.6)–(6.7) gives

$$J_0 = -1.750632969245044\ldots, \qquad H_0 = -0.137891778443756\ldots,$$

and

$$\log \pi + J_0 + H_0 = -0.743794861839401\ldots = -C$$

to the displayed precision. Identity (6.5) was checked directly at $p = 1009$ and $p = 10007$, with discrepancies -4.55×10^{-13} and 5.46×10^{-12}, respectively.

For $p = 10007$, Proposition 6.2 gives exact integer agreement at each tested cutoff:

S	3	7	15	30	60
$\mathcal{N}_p(S)$	75	119	206	374	658
right side of (6.13)	75	119	206	374	658

The corrected gamma factorization is sensitive to the direction of the Pochhammer quotients. At the same prime,

$$\log S_p \text{ (direct)} = 64659.426286523681\ldots,$$

while (6.19) gives

$$64659.426286523696\ldots,$$

a discrepancy of -1.46×10^{-11}. The reflected factor is therefore $|\sin(\pi p/q)|/\pi$, not its reciprocal.

12.3. **Exact regular-grid identity for the folded discrepancy.** Equation (5.12) was compared with a direct strict count, not with a reconstruction from the same formula. Representative values are

p	t	$H_p^<(t)$	right side of (5.12)	difference
1009	0.073	4.573630754732	4.573630754732	$-3.7 \cdot 10^{-13}$
1009	0.123	2.586401744951	2.586401744951	$3.0 \cdot 10^{-13}$
10007	0.073	4.575725433698	4.575725433686	$1.1 \cdot 10^{-11}$
10007	0.271	−3.885421385915	−3.885421385923	$8.4 \cdot 10^{-12}$
100003	0.123	−15.233017145323	−15.233017145309	$-1.4 \cdot 10^{-11}$
100003	0.271	−6.681468857360	−6.681468857427	$6.7 \cdot 10^{-11}$

At an exact jump, a computation using $\le$ instead of $<$ differs by the corresponding integer jump, as stated in Theorem 5.5. Omitting the digamma and boundary corrections leaves a bounded residual approaching $2t - 1$, in agreement with (5.13).

12.4. **A failed pure-area localized count.** The stronger heuristic

$$\#\{k \le \sqrt{p} : \|p/k\| < \delta\} \stackrel{?}{=} O(\delta\sqrt{p}) + O(1)$$

is false as a natural uniform model because divisor-rich shifts create spikes. For

$$p = 1441439, \qquad K = 1200, \qquad \delta = 2/1200,$$

the direct count is 30, while $\delta\sqrt{p} = 2.000999\ldots$. Here $p + 1 = 1441440$ has many divisors in the relevant range. No argument in this version uses the rejected pure-area estimate.

12.5. **Exhaustive folded-discrepancy maxima.** By Lemma 5.7, evaluating both one-sided values at every jump is exhaustive for the continuous supremum. On the mesoscopic interval $p^{-2/3} \le t \le 1/2$ we obtain

p	$\max \lvert H_p \rvert$	$\max \lvert H_p \rvert / p^{1/3}$	$\max \lvert H_p \rvert / (p^{1/3} \log p)$
1009	7.127365	0.710611	0.102738
5003	15.867216	0.927735	0.108917
10007	14.505816	0.673143	0.073080
20011	19.008938	0.700167	0.070695
50021	24.758911	0.671966	0.062103
100003	40.335013	0.868983	0.075479
300007	41.881577	0.625623	0.049607

The first normalized column stays between 0.62 and 0.93 on this sample, whereas division by the additional $\log p$ decreases overall. This is finite evidence for (5.15), not a proof.

At $p = 1000003$, the exact one-sided jump maxima in dyadic bands are

$[t_0, t_1)$	$\max \lvert H_p \rvert$	$[t_0, t_1)$	$\max \lvert H_p \rvert$
$[1/4, 1/2)$	44.5111	$[1/64, 1/32)$	33.8653
$[1/8, 1/4)$	40.9914	$[1/128, 1/64)$	31.6146
$[1/16, 1/8)$	41.5335	$[1/256, 1/128)$	31.2187
$[1/32, 1/16)$	44.9391	$[1/512, 1/256)$	21.1153
$[1/1024, 1/512)$	22.6831	$[1/2048, 1/1024)$	14.0273
$[1/4096, 1/2048)$	12.3144	$[1/8192, 1/4096)$	7.3583

The maximum over these displayed bands is $44.9391\ldots$, while $p^{1/3} = 100.0001\ldots$.

12.6. **Normalized prime-count scale.** The deterministic majorant entering (10.4) is

$$\sum_{p \le X} p^{-2/3} (\log p)^2.$$

Its values are

X	sum	$X^{1/3} \log X$	ratio
10^4	400.225	198.431	2.017
10^5	1181.741	534.383	2.211
10^6	3238.226	1381.551	2.344

This only checks the expected $X^{1/3} \log X$ order of the auxiliary normalization error. It does not test the von Koch remainder $\pi(X) - \mathrm{Li}(X)$.

13. Conclusion

The diagonal reciprocal sine product has two quantitatively distinct levels. The elementary block decomposition gives the density $\rho(t) = \psi_1(1 + t)$ and an $O(\sqrt{n} \log n)$ logarithmic remainder. The sharper Saffari–Vaughan discrepancy replaces this, on primes, by the proved bound

$$\log \lvert P(p) \rvert = -Cp + O(p^{1/3} (\log p)^2),$$

and propagates to the zero-omitted product, the primitive component $R_L(n) = O_\varepsilon(n^{1/3+\varepsilon})$, and absolute convergence of the associated Dirichlet series for $\Re s > 4/3$. The exact rotated Parseval identity simultaneously improves to $\|\mathscr{R}_p\|_2 \ll p^{1/3} \log p$.

The product-specific information begins after this imported discrepancy estimate. Reflection of the logarithmic sine kernel folds the full reciprocal discrepancy into $H_p(t) = D_p(t) - D_p(1 - t)$. The exact Stieltjes formula (5.4) shows that the extreme singular range contributes only $O(p^{1/3} \log p)$; the second logarithm in the current pointwise theorem can arise only in the mesoscopic range $p^{-2/3} \le t \le 1/2$. The exact sawtooth identity (5.12) puts this folded object on a regular reciprocal grid, including the digamma and boundary corrections that are numerically essential. Exhaustive finite computations support, but do not prove, the stronger bound $H_p(t) \ll p^{1/3}$, which would remove one logarithm from the product remainder.

The singular logarithm also separates into a bounded-variation term and the integer core $S_p = \prod s_k$. The latter is an exact statistic of the symmetric shifted divisors of $p - s$ and $p + s$, admits the corrected gamma factorization (6.19), and has a low-complementary-divisor sector that can be smoothed to precision $O(p^{1/3} \log p)$. Gamma reflection alone does not close the remaining pointwise problem.

The later Euclidean Lambert, Clausen, cyclotomic, Möbius, totient, and Chebyshev identities remain exact structural reorganizations. The improved prime normalization now satisfies

$$S_c(X) = \frac{1}{2}\pi(X) + O(X^{1/3} \log X),$$

but this sharper auxiliary approximation is still absorbed inside the classical von Koch scale and therefore does not yield new information about the zeros of ζ. The unresolved mathematical problem specific to the present product is the mesoscopic folded discrepancy, not the Riemann hypothesis.

Figures 1–5 summarize this hierarchy for a reader encountering the product for the first time: the raw zero/nonzero product and its continuous envelope $x \mapsto e^{-Cx}$ treated explicitly as a function of a real variable, the zero-omitted repair onto a single exponential line, the direct numerical convergence $-\log|P(p)|/p \to C$, the chain of auxiliary objects ρ, ℓ, K_J, D_p, H_p through which the proved $p^{1/3}(\log p)^2$ scale and the open $p^{1/3}$ conjecture are stated, and finally the centered kernel V_k, the Clausen function, and the bounded-variation kernel h used in the later sections. Remark 5.6 also records, for the historical record, the exact three-line reason a linear (non-reciprocal) variant of (5.12) carries no arithmetic content whatsoever: it is identically $2t-1$, independent of p and of primality, and cannot be a restatement of the folded discrepancy without asserting the false claim $H_p = O(1)$. This version closes the present line of investigation: every theorem above is proved unconditionally, every numerical claim is independently reproducible from the displayed identities, and the single explicit open question, inequality (5.15) for the folded discrepancy on the mesoscopic range, is stated as a conjecture and nothing more.

References

[1] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, New York, 1976. doi:10.1007/978-1-4757-5579-4.

[2] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Mathematical Society Lecture Note Series 126, Cambridge University Press, Cambridge, 1991. doi:10.1017/CBO9780511661976.

[3] S. Grepstad, L. Kaltenböck, and M. Neumüller, *On the asymptotic behaviour of the sine product* $\prod_{r=1}^{n} |2\sin(\pi r\alpha)|$, arXiv:1909.00980 (2019). https://arxiv.org/abs/1909.00980.

[4] G. H. Hardy, J. E. Littlewood, and G. Pólya, *Inequalities*, 2nd ed., Cambridge University Press, Cambridge, 1952.

[5] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, Oxford, 2008.

[6] J. Herrera-Poyatos and P. Moree, *Coefficients and higher order derivatives of cyclotomic polynomials: old and new*, MPIM Preprint 2018 (46). https://archive.mpim-bonn.mpg.de/1354/.

[7] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, New York, 1981.

[8] X. Li and X. Yang, *An improvement on Gauss's circle problem and Dirichlet's divisor problem*, arXiv:2308.14859 (2023), preprint. https://arxiv.org/abs/2308.14859.

[9] D. H. Mayer, *Continued fractions and related transformations*, MPIM Preprint 1989 (73). https://archive.mpim-bonn.mpg.de/1282/.

[10] C. O'Sullivan, *Asymptotics for the partial fractions of the restricted partition generating function I*, International Journal of Number Theory **12** (2016), 1421–1474. doi:10.1142/S1793042116500895.

[11] MikeTeX, *Distribution of the fractional parts of* $n/1, n/2, \ldots, n/n$ *as* n *tends to infinity*, Mathematics Stack Exchange, question 4831482, 21 December 2023. https://math.stackexchange.com/questions/4831482/.

[12] B. Saffari and R. C. Vaughan, *On the fractional parts of* x/n *and related sequences. II*, Annales de l'Institut Fourier **27** (1977), no. 2, 1–30. doi:10.5802/aif.649.

[13] I. J. Schoenberg, *Metric spaces and positive definite functions*, Transactions of the American Mathematical Society **44** (1938), 522–536. doi:10.1090/S0002-9947-1938-1501980-0.

[14] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.

[15] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Graduate Texts in Mathematics 83, Springer, New York, 1997. doi:10.1007/978-1-4612-1934-7.

[16] C. P. Willans, *On formulae for the nth prime number*, The Mathematical Gazette **48** (1964), no. 366, 413–415. doi:10.2307/3611701.

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